Chapter 2

Groups, First Encounter

2.1 Definition of a Group

Exercise 2.1.1. Write a careful proof that every group is the group of isomorphisms of a groupoid. In particular, every group is the group of automorphisms of some object in some category.

> **Solution.** Let G be any group. It suffices to construct such a groupoid such that the collection of isomorphisms is isomorphic to G. Let $\mathrm{Obj}(\mathsf{G}) = \{*\}$. For all $g \in G$, let there exist a morphism $f_g \in \mathrm{End}_{\mathsf{G}}(*)$, and let morphism composition be defined as
>
> $$f_g \circ f_h = f_{gh},$$
>
> for $g, h \in G$, using the notation as previously. To show this is a category, it suffices to show there is an identity morphism. Note that f_e is clearly an identity, since $f_g \circ f_e = f_e \circ f_g = f_g$. Therefore, G is a category. Now we show G is a groupoid. Let $f_g \in \mathrm{Hom}_{\mathsf{G}}(*, *)$. Since $g \in G$, there exists $g^{-1} \in G$ such that $gg^{-1} = g^{-1}g = e$. Equivalently, there exists $f_{g^{-1}} \in \mathrm{End}_{\mathsf{G}}(*)$ such that
>
> $$f_g \circ f_{g^{-1}} = f_{g^{-1}} \circ f_g = f_e.$$
>
> Therefore, all morphisms have inverses, and G is a groupoid. Since G has only one element, the collection of (iso)morphisms in G is precisely $\mathrm{Aut}_{\mathsf{G}}(*)$, which we have shown to be a group already. Let $\phi G \to \mathrm{Aut}_{\mathsf{G}}(*)$ be the map $g \mapsto f_g$. Clearly the map is well-defined and surjective. Also note that
>
> $$\phi(gh) = f_{gh} = f_g \circ f_h = \phi(g)\phi(h).$$
>
> Therefore, ϕ is a homomorphism, and we need only show ϕ is injective. Suppose $\phi(g) = \phi(h)$. That is, $f_g = f_h$. Since inverses are unique, there exists exactly one inverse f_a to the above morphisms. Since the identity is unique, we must have $ag = ga = e$ and $ah = ha = e$. Cancelling a, we have $g = h$. ■

Exercise 2.1.2. Consider the sets $\varnothing, \mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ under $+$ and $\cdot$. Which are groups, and which aren't?

> **Solution.** Clearly $\mathbb{N}$ is a group under none since it isn't closed under addition and has no multiplicative inverses. $\mathbb{Z}$ is a group under addition but not under multiplication. $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are all groups under $+$, and are groups under $\cdot$ if zero is removed. ■

Exercise 2.1.3. Prove that $(gh)^{-1} = h^{-1}g^{-1}$ for all $g, h \in G$ a group.

Solution. Let $g, h \in G$. Note that

$$(gh)\left(h^{-1}g^{-1}\right) = g\left(hh^{-1}\right)g^{-1} = ge_G g^{-1} = gg^{-1} = e_G.$$

Therefore, $(gh)^{-1} = h^{-1}g^{-1}$. ■

Exercise 2.1.4. Suppose $g^2 = e$ for all $g \in G$ a group. Prove G is abelian.

Solution. Let $g, h \in G$. Since $g^2 = e$ for all $g \in G$, we have $g = g^{-1}$ for all $g \in G$. By the previous exercise, $(gh)^{-1} = h^{-1}g^{-1}$. However, since $g = g^{-1}$ for all $g \in G$, we have

$$gh = hg.$$

■

Exercise 2.1.5. Prove every row and column of the Cayley table for a group contains each element exactly once.

Solution. It suffices to show that left- and right-multiplication by a group element is a bijection $G \to G$. Let $g \in G$, and consider the map $a \mapsto ga$. Clearly this map is well defined, and it has an inverse map $a \mapsto g^{-1}a$, so it is a bijection. Right multiplication follows by the same logic. ■

Exercise 2.1.6. Prove there is only one multiplication table for groups with 1, 2, or 3 elements. Show there are two tables for groups with 4 elements.

Solution. The case for 1 is trivial. For the second case, let $\{e, g\}$ be a group. We must have $eg = ge = e$, $ee = e$, and since g needs an inverse, we must have $gg = e$. Therefore, the group requirement forces this unique multiplication structure.

Now let $\{e, a, b\}$ be a group. Suppose for contradiction that a, b are not inverses. Then ab is either a or b. The inverse of a must then be a, and likewise for b. It follows that $e = a^2 = aba$. Cancelling a once on the left and right gives us $a^2 = e = b$, a contradiction. Therefore, $a = b^{-1}$, and this forces a unique structure.

I have done this proof before and I don't feel like repeating it so I'm just going to leave it here. ■

Fuck this

2.2 Examples of Groups

2.3 The Category Grp

Exercise 2.3.1. Let $\phi : G \to H$ be a morphism in a category C with products. Explain why there is a unique morphism $(\phi \times \phi) : G \times G \to H \times H$ compatible in the evident way with the natural projections.

Solution. I think it's good to spell out what the "evident way" is. We want the diagram

$$\begin{tikzcd}
 & G \arrow[r, "\phi"] & H \\
G \times G \arrow[ur, "\pi"] \arrow[r, "\phi\times\phi"] \arrow[dr, "\pi"] & H \times H \arrow[ur, "\pi"] \arrow[dr, "\pi"] & \\
 & G \arrow[r, "\phi"] & H
\end{tikzcd}$$

to commute. Note that ϕ determines a map $G \times G \to H$ given by $\phi\pi$. By the universal property of products, there exists precisely one morphism $\sigma : G \times G \to H \times H$ such that the diagram

$$\begin{tikzcd}
 & & H \\
G \times G \arrow[urr, bend left, "\phi\pi"] \arrow[r, "\sigma"] \arrow[drr, bend right, "\phi\pi"] & H \times H \arrow[ur, "\pi"] \arrow[dr, "\pi"] & \\
 & & H
\end{tikzcd}$$

commutes. Note that these diagrams are the same, except $\phi \times \phi$ has been replaced by σ. We can take this unique morphism σ to be the morphism $\phi \times \phi$. ■

Exercise 2.3.2. Let $\phi : G \to H$, $\psi : H \to K$ be morphisms in a category C with products, and consider morphisms between the products $G \times G$, $H \times H$, and $K \times K$. Prove that

$$(\psi\phi) \times (\psi\phi) = (\psi \times \psi)(\phi \times \phi).$$

Solution. For simplicity, since the diagrams are symmetric, I will only draw the one side of them. We have

$$\begin{tikzcd}
G \times G \arrow[r, "\phi\times\phi"] \arrow[d, "\pi"] & H \times H \arrow[r, "\psi\times\psi"] \arrow[d, "\pi"] & K \times K \arrow[d, "\pi"] \\
G \arrow[r, "\phi"] & H \arrow[r, "\psi"] & K
\end{tikzcd}$$

Note that by previous logic, $(\psi \times \psi)(\phi \times \phi)$ is a unique morphism $G \times G \to K \times K$. It suffices to show then that $(\psi\phi) \times (\phi\psi)$ exists, and is a morphism $G \times G \to K \times K$. This follows fairly immediately. Condider the commutative diagram

$$\begin{tikzcd}
G \times G \arrow[r, "\exists!\sigma"] \arrow[d, "\pi"] & K \times K \arrow[d, "\pi"] \\
G \arrow[r, "\psi\phi"] & K
\end{tikzcd}$$

Since the products are symmetrical, this is in fact the full diagram for the universal property of products. By the same logic then, we have $\sigma = (\psi\phi) \times (\psi\phi)$. ■

Exercise 2.3.3. Show that if G, H are abelian groups, then $G \times H$ satisfies the universal property for coproducts in Ab.

Solution. Recall the universal property of coproducts

$$\begin{tikzcd}
G \arrow[dr, "i"] \arrow[drr, bend left, "\phi"] & & \\
& G \times H \arrow[r, "\exists!\sigma"] & Z \\
H \arrow[ur, "i"] \arrow[urr, bend right, "\psi"'] & &
\end{tikzcd}$$

Consider $\sigma(a, b) = \phi(a) + \psi(b)$. Note that

$$\sigma i_G(a) = \sigma(a, 0) = \phi(a) + 0_G = \phi(a),$$

and similarly for i_H. Therefore, the diagram commutes. Showing the morphism is unique is more difficult, and will be done at the end. Now we will show σ is a homomorphism. Let $a, c \in G$ and $b, d \in H$. We have

$$\sigma(a, b) + \sigma(c, d) = \phi(a) + \psi(b) + \phi(c) + \psi(d) = \phi(a + c) + \psi(b + d) = \sigma((a, c) + (b, d)).$$

Note that this uses the abelian hypothesis, explaining why this does not work in Grp. Finally, we show the map is unique. We must have $\sigma(a, 0) = \phi(a)$ and $\sigma(0, b) = \psi(b)$ for all $a \in G$, $b \in H$ due to the inclusion map definitions. Therefore,

$$\sigma(a, 0) + \sigma(0, b) = \phi(a) + \psi(b).$$

Using the homomorphism condition, we have

$$\sigma(a, b) = \phi(a) + \psi(b).$$

Therefore, assuming any other definition would cause it to fail. ■

Exercise 2.3.4. Let G, H be groups, and assume that $G \cong G \times H$. Can you conclude that H is trivial?

Solution. h ■